

CHEATSHEET · QUICK REFERENCE

Lab 10 — Server Roles: Web, File, and Database

Stand up the three most-tested roles (nginx web, Samba file, MariaDB database) end-to-end, isolate them with systemd resource limits and document the inventory.

Web role (nginx)

```
systemctl start nginx
Start the web server

curl -s http://127.0.0.1/ | head
Confirm HTTP response

nginx -t && systemctl reload nginx
Validate + reload a vhost
```

File role (Samba)

```
(echo 'P@ssw0rd!'; echo 'P@ssw0rd!') | smbpasswd -s -a alice
Create an SMB user

chmod 2770 /srv/share
Set-GID group share permissions

smbclient //127.0.0.1/share -U alice%P@ssw0rd! -c 'ls'
Test the share
```

Database role (MariaDB)

```
mysql -e "CREATE DATABASE appdb;"
Create the database

mysql -e "CREATE USER 'app'@'localhost' IDENTIFIED BY 'AppPass!1';"
Dedicated service account

mysql -e "GRANT ALL ON appdb.* TO 'app'@'localhost';"
Least-privilege grant

systemctl set-property mariadb.service MemoryMax=512M CPUQuota=80%
Resource limits
```

Per-role checklist

Role	Port	Key controls
Web	80 / 443	docroot perms, logs, TLS
File	445	auth backend, ACLs, quotas, audit
Database	3306	service account, least-privilege, separate data disk